

Cancellations of Limit Orders

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Abstract

More than 20% of all the orders submitted to the NYSE SuperDot are cancellation orders and 40% of all the limit orders submitted are cancelled. This paper investigates into the motivation behind these cancellations. Three trading strategies which involve cancellations account for up to 85% of all the cancellation orders. Subsequent to cancellations, limit order traders are more likely withdraw from trading, or resubmit the limit orders with more aggressive limit prices, than to replace the cancelled orders with market orders. The paper also shows that the decision of traders to cancel their orders is influenced by market and order characteristics.

Introduction

Order cancellations have not received much attention in the finance literature. One reason may be the lack of data to study the nature and impact of cancellations. Another more important reason may be that there were not many order cancellations in the past. Data from the New York Stock Exchange (NYSE) SuperDOT in 1990 (the TORQ data set) shows that only about five percent of the total number of orders submitted were cancellation orders. However, the situation has changed drastically over the last decade. New data from the NYSE SuperDOT collected in 2001 (used in this paper) shows that cancellation orders typically constitute more than 20% of all the orders submitted in any particular day. In fact, there were more cancellation orders than there were market buy or market sell orders. Almost all of the cancellation orders were limit order cancellations. More than 40% of all limit orders submitted were cancelled. This paper investigates the reasons behind the massive cancellations that are observed in the data and the variables that may influence a trader's decision to cancel his orders.

Cancellations are important especially in limit order trading. On one hand, limit orders that sit in the queue, while the prices move further and further away, need to be cancelled and resubmitted at more aggressive prices, if the traders wish to trade. On the other hand, new information may enter the market during the time that the limit orders sit in the queue. The ability to cancel these stale limit orders in time is crucial to prevent them from being executed at unfavorable prices. The inability to cancel stale orders in time may have huge and adverse consequences. Kleidon (1992) finds that long queues at the printers at the specialists' posts, during the crash of October 1987, resulted in long delays

between order submission and execution. This coupled with the failure of the software designed to allow cancellations or replacement of limit orders, resulted in large number of stale limit orders being executed, which in turn resulted in the delinkage between the S&P 500 index's cash and future market.

One of the very few papers that touch on order cancellations is Biais, Hillion and Spatt (1995). They find a positive serial correlation in the submission of order cancellations in the Paris Bourse. In other words, the cancellations follow each other quickly. They find that the cancellations at the bid (ask) are particularly frequent after market buy (sell) orders. This may be due to traders using market orders to test for hidden orders in the book and then cancel the orders in the absence of hidden orders. Their results show that cancellations do not vary with the spread. Ellul, Holden, Jain and Jennings (2003) also finds a positive serial correlation in the submission of cancellation orders. Their results show that cancellations are more likely with wide spreads (contrary to Biais, Hillion and Spatt (1995)) and that the number of cancellation orders submitted decreases as the market closing time draws near. The paper looks at the conditions under which the observation of a cancellation order is most likely. My paper emphasizes, instead, on the reasons behind the submission of these cancellations.

Besides understanding the reasons behind the submission of cancellation orders, this paper looks into what the traders do after they cancel their orders. In an attempt to theorize the rationale for limit order trading, Handa and Schwartz (1996) supposes that if a limit order failed to be executed, it will either be withdrawn (and not trade) or be

replaced by (i.e. cancelled and resubmitted with) a market order at the end of a trading window. However, the paper do not allow for a third possibility: that the limit order is cancelled and replaced by a more aggressively priced limit order. In a paper comparing the performance between market orders and limit orders, Harris and Hasbrouck (1996), attributes an opportunity cost to an unexecuted limit order by assuming that the unexecuted or cancelled limit orders are filled at the opposite-side quote prevailing at the time of cancellation. This is essentially equivalent to assuming that a cancelled limit order is immediately replaced by a market order. Once again, the case of replacing a cancelled limit order by another limit order is not considered.

I examine two trading strategies that might contribute to the large number of cancellations in the data. The first is the order splitting strategy. A limit order trader wishing to disguise his large buying or selling interest may break up his order into smaller ones. If, for example, an order is split into five smaller orders, it will need five cancellation orders to cancel all of them instead of one. The second strategy is one of undercutting. Undercutting is the behavior when more than one trader submits limit orders to gain price priority over one another within a short time interval. Traders with unsuccessful undercutting orders may cancel their orders and either engage in further undercutting by resubmitting these orders or leaving the market altogether.

Using data from the NYSE, I find that up to 71.49% of the cancellations are cancellations of split orders; while up to 24.59% of the cancellations are cancellations of unsuccessful undercutting limit orders. I also find that most traders withdraw from trading in the stock

subsequent to cancellation. If however, they choose to continue to trade, they are much more likely to resubmit their orders with more aggressively priced limit orders rather than market orders. Less than 2% of the cancelled limit orders are replaced by market orders, whereas as many as 34.19% of them are replaced by limit orders. Combining all the three trading strategies, namely order splitting strategy, undercutting strategy, and limit order replacement strategy, I am able to explain up to 85.03% of all the cancellations in the sample.

Controlling for order splitting and undercutting behavior among the limit order traders, the paper continues to analyze variables that may influence a trader's decision to cancel his limit orders after submitting them. The probability of cancellation of a limit buy (sell) order is higher when there is a positive (negative) change in the bid (ask) after the order is submitted. There is a positive relationship between the cancellation of a limit order and the change of the spread after order submission. The likelihood of a limit order being cancelled is higher as the trading day proceeds. However, the likelihood is lower at the last half hour of trading. Cancellation of a limit buy (sell) order is also more likely during rising (falling) stock prices.

The organization of the paper is as follows. The data and some descriptive statistics are described in section I. Section II looks at how the order splitting strategy and undercutting strategy contribute to the large number of cancellation orders. Section III investigates into the actions of traders subsequent to order cancellations. Section IV combines the trading strategies examined in this paper to explain the cancellations in the

sample. Section V presents the probit regression analysis of the variables that may influence the traders' decision to cancel their orders. Section VI concludes the paper.

I. Data and Descriptive Statistics

The data set used in this paper is similar to the TORQ data compiled by Joel Hasbrouck (Hasbrouck (1992)). This data set consists of the System Order Database (SOD) and its companion quote file (SODQ). This new data set will be collectively referred to as the SOD (2001) henceforth. This data set comprises 148 securities for the period from April to June of 2001. There are two fields (not present in the TORQ (1990) data) in the SOD (2001) that are instrumental to the identification of the trading strategies examined in this paper. One field gives a unique code to each brokerage firm, and another field gives a unique code to each branch of each of the brokerage firm, who enters an order to the SuperDOT system. Due to the massive size of the data, only one day, April 11, 2001, is sampled. Another day, April 2, 2001, is used as a robustness check. The results on both days are consistent. Only pure market and limit orders during the NYSE official market opening time are considered. Harris and Hasbrouck (1996) show that of all the SuperDot orders, pure market orders and limit orders made up 95 percent of the all orders. The SOD (2001) data shows a similar trend. In the sample period of this paper, straight market orders and standard limit orders account for about 98 percent of the orders. Table I shows the distribution of the different types of orders. Of the 744, 969 orders submitted in the sample, 12.14% and 14.27% of them are market buy and market sell orders respectively. Limit buy and limit sell orders constitute 27.17% and 24.18% of the total number of orders respectively ^[1]. There are 165,720 cancellation orders constituting

22.25% of all the orders submitted. It is indeed surprising to note that the number of cancellations is more than the number of market buy or sell orders.

The breakdown of the cancellation orders are recorded in Table II. Of all the cancellations in the sample, only 1.73% of them are market order cancellations, while 98.27% of them are cancellation of limit orders. For this paper, only limit order cancellations will be studied. In order to analyze these cancellation orders, they have to be first matched to their corresponding originating orders. For a cancellation order to be matched to its originating order, they must both be for the same stock, submitted by the same branch of the same firm, have the same turnaround number, of the same type of order (market or limit), and have the same order size. I only manage to match 89.39% of all the limit order cancellations. The analysis of this paper will only be performed on these matched limit order cancellations.

Panel C of Table II shows the break down of these matched cancellations. 52.01% of them are for limit buys, while 47.99% of them are for limit sells. Panel D further breaks down the cancellations into whether there is a positive, negative, or no change in the respective best quotes between the time of the submission of the original orders and their cancellations. As expected, most of the cancellation occurs when the quotes are moving away, i.e. a positive change in the bid for limit buys and a negative change in the ask for a limit sell. However, there are still a substantial number of cancellations whereby there is either no change in the respective quotes or when there are favorable movements in the quotes.

Table III shows the distribution of the time interval between the submission of the limit orders and their cancellations. It is surprising to see that 21.50% of the cancellations are submitted within 10 seconds of the submission of the original orders. 61.92% of the cancellations are submitted within a minute of the submission of the original orders, and 95.49% of the cancellations are submitted within 10 minutes after their original orders.

II. Order Splitting Strategy and Undercutting Strategy

A. Order Splitting Strategy

Larger orders tend to have larger market and price impact. Large limit orders take longer to fill and expose the trader to adverse price moments and undercutting. One way a trader can disguise his large buying or selling interest is to split up the order into smaller ones and submit them, either all at one time or, one after another. It is not difficult to imagine why such an order splitting strategy may be one of the causes of the large number of cancellations in the sample. If an order is split into five smaller orders, it will need five cancellation orders to cancel all of them instead of one. I define order splitting as orders that are of the same type (market or limit order), in the same direction (buy or sell), for the same stock, that originated from the same branch of the same firm, and that are separated by a short time interval (10, 30, 60, 120, and 300 seconds). The cancellation orders are checked to see if their originating orders are split orders. The results are shown in Table IV.

As expected, order splitting does play a major role in contributing to the number of cancellation of limit orders. For the full sample, the percentage of the originating limit orders that are split orders increases from 12.67% to 71.49% as the time window increases from 10 seconds to 5 minutes (300 seconds). I divide the full sample into 4 sub-samples according to trading activity: Quartile 1 contains the most active stocks and Quartile 4 contains the least active ones. Similar to the full sample, the percentage of order splitting in the originating orders increases as the time window increases. Cancellations due to order splitting are more common for active stocks than less active ones.

A limit order trader can use order splitting as a strategy to reduce his trading risk. The strategy allows him to disguise his large interest and may reduce the price impact of his trade. At the same time, since smaller orders are easier and quicker to fill than larger ones, the strategy may reduce the time his orders sit in the queue and thus reduce the exposure risk of his orders. The ability to cancel orders, when cancellation is optimal, is crucial for a trader adopting an order splitting strategy. His willingness to split his orders, and the number of orders he is willing to split his orders into, is closely tied to the ability of the exchange to efficiently handle order cancellations. Delays in cancellation may result in orders being executed at unfavorable prices. Consider a case whereby a trader splits his order into five smaller orders. He subsequently wishes to cancel all five orders due to some new information that has just entered into the market. If the exchange can only handle two cancellations in time, the three remaining orders will be subjected to unnecessary risk. If he has prior knowledge of such possible delays, he may choose to

split his order into lesser number of orders or he may not be willing to split his order at all. Traders who take for granted the ability to cancel orders as and when they desire should learn a lesson from the October 1987 crash as discussed in Kleidon (1992) and mentioned in the introduction above. Hence, though not always obvious, cancellation is an important part of an order splitting strategy.

B. Undercutting Strategy

The more aggressively the limit orders are priced, the higher the probability that they will be executed. Undercutting is the behavior when more than one trader submits limit orders to gain price priority over one another within a short time interval. We should expect undercutting to be very common in the NYSE, especially after decimalization. Limit order traders can undercut the best quotes (and thereby undercutting one another) by improving the quotes by as little as one penny. Some attempts to undercut may be successful, resulting in trades, while others may not be. Traders with unsuccessful undercutting orders may cancel their orders and either engage in further undercutting by resubmitting these orders or leaving the market altogether. I define undercutting to be the case when a limit order improves on the current quote, within a short time interval (10, 30, 60, 120, and 300 seconds), after it has just been improved upon by another limit order. The originating orders of the cancellations are checked to see whether they are undercutting orders. Table V shows the results.

Cancellation of undercutting limit orders constitutes 19.21% of all limit order cancellations when the time window is set to 10 seconds. This percentage increases to

24.59% as the time window increases to 5 minutes (300 seconds). Although the cancellation of undercutting limit orders as a percentage of all limit order cancellations increases with the length of the time window allowed between two undercutting orders, the increase is much more gradual than the that which we see in the case of order splitting. The quartiles show similar results. In comparison with the order splitting strategy, for the full sample and the two larger quartiles (more active stocks), undercutting behavior among limit order traders contributes less to the total number of cancellations. However, the opposite is true for the two smaller quartiles (less active stocks).

As with the case of the order splitting strategy, order cancellation is an important part of an undercutting strategy. Traders must be allowed to cancel their orders timely in order for such a strategy to be successful. An efficient cancellation system is especially essential to allow for multiple players to undercut one another in a multi-stage undercutting game, where each player can choose one of three actions at each stage: leave his order unchanged, cancel and exit the game, or cancel and resubmit. Two of the three choices involve order cancellations. The results in this section indeed show that limit order traders do use cancellation as a means to achieve their undercutting objective.

III. Actions Subsequent to Cancellations

A trader can choose one of three courses of action after canceling his order: withdraw totally, replace with a market order, or replace with a more aggressively priced limit order. Papers in the literature, such as Handa and Schwartz (1996) and Harris and

Hasbrouck (1996), consider the first two courses of action. All three options are considered in this section.

A. Replacement with Market Orders

Orders submitted by the same branch office of the same member firm, for the same stock, within a short time interval are considered to be orders submitted by the same trader. The advantage of choosing a short time interval is two-fold. Firstly, it is highly likely that the trader would want to replace his cancelled orders (if he wants to replace them) as soon as possible. Secondly, the shorter the time interval is, the higher the probability that the orders are submitted by the same trader. Hence, for this section, we adopt the same time windows as those that were used in the previous sections. A trader is considered to have replaced his limit buy (sell) order by a market buy (sell) order, when a particular branch of a particular firm, submits a market buy (sell) order, for a particular stock, within a short time window, after a limit buy (sell) order, for the same stock, submitted by the same branch of the same firm, has been cancelled.

Table VI shows the results. A trader who wishes to replace his cancelled limit orders with market orders is likely, but not constrained, to keep the number of shares unchanged. Hence, two scenarios are considered. The first case is when the order size for the replacement market order is the same as the cancelled limit order (Panel A); and the second case is when this constraint is relaxed (Panel B). The first case is a subset of the second, and the true figures should be some where in between. As we can see from the results, the practice of replacing cancelled limit orders with market orders is not

commonly seen in the data. In Panel A, the case of equal order size, for a 10-second time window, only 0.11% of all the cancelled limit orders are replaced by market orders. This proportion increases to less than half a percent point (0.49%) at the longest time window considered. The replacing of limit orders by market orders is even less common in less active stocks. As we go from the first to the fourth quartile, the proportion of cancelled limit orders replaced by market orders falls. For the least active quartile, there is no evidence that any of the cancelled limit orders is replaced by market orders. Though the figures are greater in Panel B, when the constraint of equal order size is relaxed, such replacement practice is still rather rare. None of the percentages, whether in the full sample or in the quartiles, is greater than 2%. Still none of the cancelled limit orders in the least active quartile is replaced by market orders. Hence the data do not show support that subsequent to canceling their limit orders, traders tend to resubmit them as market orders.

B. Replacement with Limit Orders

One thing that both Harris and Hasbrouck (1996) and Handa and Schwartz (1996) do not consider, is the possibility that the cancelled limit orders re-enter the book with more aggressive limit prices. In other words, limit order traders actively replace limit orders, with low probability of execution, with those that have higher probability of execution. Table VII shows the number of cancellations that are replaced with more aggressively priced limit orders. Once again, these replacement limit orders are not constrained to have the same order size. Panel A shows replacement limit orders with the same order size as the original orders, while Panel B relaxes the constraint.

Replacing cancelled limit orders with more aggressively priced limit orders is much more common than with market orders. In Panel A, allowing a 10 seconds interval between order cancellation and resubmission, 7.22% of all the cancellations are replaced with more aggressive limit orders. This percentage increases to 15.88% as we increase the time window to 300 seconds. Although replacing limit orders with better limit orders is generally more common for actively traded stocks, the difference is not great. For the 300-second window, for example, 16.00% of all the limit order cancellations are replaced by better limit orders for Quartile 1; whereas the percentage is 12.61% for Quartile 4. The general results in Panel A are repeated in Panel B except that the percentages are much higher. In some of the cases, the proportion of cancellations that are replaced by better limit orders are more than double. The difference between the more active stocks and the less active ones are also larger.

We can now draw some conclusion regarding the action of limit order traders following cancellation. As mentioned earlier, limit order traders have three courses of action following cancellation; namely, withdraw from further trading, resubmitting as a market order, or resubmitting as a more aggressively priced limit order. The results of this section show that if a limit trader do replace his cancelled limit orders, he is much more likely to replace them with more aggressively priced limit orders than with market orders. Since a combination of these two mutually exclusive actions explains less than 50% of all the cancellations, it implies that most of the cancelled limit orders are not replaced at all. In other words, in most cases, limit order traders exit the market altogether after canceling their orders ^[2].

Harris and Hasbrouck (1996) qualifies rightly that, to assume that all expired or cancelled limit orders are immediately resubmitted as market orders will exaggerate the penalty associated with limit order execution failure. As evidenced in this section, most of the limit traders have low urgency to trade. Even if they are pre-committed to trade, many do so by replacing their failed limit orders with other limit orders, and so do not bear the full cost of immediately trading with market orders.

IV. Trading Strategies with Cancellations

Although we concluded, in the previous section, resubmission of more aggressively priced limit orders as a common subsequent action to limit order cancellations, such cancellation and subsequent resubmission is, to a large extent, a trading strategy. Obviously, cancellation is a vital part of such a limit order replacement strategy. The limit order replacement strategy may be (but not necessarily) a follow-up strategy to the undercutting strategy. When a limit order trader cancels his unsuccessful undercutting limit orders, he may choose not to exit the undercutting game, but re-enters the game by resubmitting his limit orders with more aggressive prices. However, not all cancelled undercutting orders are replaced and not all replaced orders are for failed undercutting orders.

Hence, thus far, we have analyzed three different trading strategies in which limit order cancellation plays an important role: namely, the order splitting strategy, the undercutting strategy, and the limit order replacement strategy. We shall now examine the total

contribution, of all the three strategies combined, to the cancellations in the sample. Table VIII shows the percentage of the total number of cancellations that are caused by either one or more of the three trading strategies. Two scenarios of the limit order replacement trading strategy are considered. Panel A shows the case whereby the replacement limit orders are of the same order size as the original orders, while Panel B shows the case whereby it may be different.

In Panel A, the three strategies contribute 33.78% of the total number of cancellations in the sample if we consider a time window of 10 seconds. The percentage increases to 81.96% if we allow a time window of 300 seconds. Such trading strategies are more common in active stocks. The percentage of cancellations explained decreases as we go down the quartiles from the most active one (Quartile 1) to the least active one (Quartile 4). For the 300-second window, the strategies are able to explain more than half of all the cancellations for the three more active quartiles, and 41.18% of the cancellations for the least active one. Such trends among the quartiles are mirrored in Panel B. The percentages are higher (across the board) in Panel B than in Panel A, though the differences are not large. The proportion of the cancellations caused by the three trading strategies increases from 38.18% to 85.03% as the time window increases from 10 seconds to 5 minutes. In other words, this paper is able to explain up to 85.03% of all the cancellations in the sample.

V. Probit Analysis of Variables Influencing Cancellation Decisions

Although up to 85 percent of all the cancellations can be accounted for, it is still useful to learn how market and order characteristics may influence a trader's decision to cancel his orders. In this section, I use a series of probit regressions to analyze the variables that may have an effect on the probability that a limit order will be cancelled. This section is different from that of Ellul, Holden, Jain and Jennings (2003) in several aspects. Firstly, Ellul, Holden, Jain and Jennings (2003) considers the probability of observing, among other types of orders (e.g. market orders and limit orders of different limit prices), a cancellation order. However, this section considers the probability that a limit order, after it is submitted, will be cancelled. Secondly, Ellul, Holden, Jain and Jennings (2003) controls for order splitting in their study. I will control for both order splitting and undercutting in my analysis. Lastly, Ellul, Holden, Jain and Jennings (2003) considers the probability of observing a cancellation at one point in time. I consider the probability of cancellation of the limit orders at different time intervals after they are submitted.

All 382,561 limit orders (see Table I) are considered in this analysis. Split orders and undercutting limit orders are removed from the sample in order to control for these two effects. The 60-second time window is used to determine order splitting and undercutting. The decision to cancel an order, after it is submitted, is an on going process, ending when the order is finally terminated either by execution or cancellation. At every instant in time, before the order is terminated, the trader is making an active decision to leave the order alone or to cancel it. However, analyzing such a dynamic-on-going decision process is difficult, if not impossible. Table III shows that most of the cancellations occur

between 10 to 300 seconds after the orders are submitted. We can also see that almost all (95.49%) of the cancellations occur within 600 seconds of submission. Hence, I will model the decision to cancel at 6 different time intervals after the orders are submitted: namely, 10, 30, 60, 120, 300, and 600 seconds. For each time window (for example the 60-second time window), limit orders that are either executed or cancelled before that the particular time window (i.e. 60 seconds after submission) will be deleted and the remaining existing limit orders are analyzed.

I analyze the cancellation of limit buy and limit sell orders separately. The direction of the change of the bid and the offer quotes, from the moment a trader submits his limit orders, is likely to have an impact on his decision to cancel his orders. An increase (decrease) in the bid (offer) reduces the probability of execution for a limit buy (sell) order. The variables *cngbid* and *cngofr* are dummy variables indicating the direction of the change of the bid and offer quotes respectively. *Cngbid* (*cngofr*) takes the value of 1, if there is a positive (negative) change in the bid (offer) price, and 0 otherwise. Hence, the coefficient for *cngbid* (*cngofr*) should be positive for a limit buy (sell) order. The *ordsz* variable measures (in thousands of shares) the size of the order. The larger the order size, the longer it will take to fill the order. Hence, larger orders are exposed to more risk, and therefore are more likely to be cancelled. We will therefore expect a positive relationship between order size and order cancellation. The bid-offer spread is positively related to the volatility of a stock. The *spread* variable in the regression measures the percentage change in the bid-offer spread within the respective time window. A positive change in the spread indicates an increase in the volatility of the

stock. Higher volatility may imply more frequent revision of the valuation of the stocks by the traders, and hence increases the probability of cancellation of the limit orders. The sign on this coefficient should, therefore, be positive. I use two different variables to test the effect of the time on the cancellation of limit orders. The *otime* variable measures the time of the day (in hours), after the market opening time, when the order is submitted. The *atime* variable is a dummy variable that takes the value of 1 if the order is submitted half an hour before the market official closing time, and 0 otherwise. I expect *otime* to be positively related to limit order cancellation for two reasons. First, as the trading day proceeds, the time left for the limit orders to be executed shortens, and thus the probability of execution, everything else constant, decreases. Second, for a pre-committed trader who must trade by the end of the day, his patience will decrease with the time left to trade. On the other hand, for the last half hour of a trading day, since there is not much time left to trade, traders may tend to be more careful with the pricing of their limit orders to ensure an execution if they desire one. Hence, the sign on *atime* should be negative. The *cngpx* variable measures the percentage change in the trading price of the stock during the respective time window. As the price of the stock changes, the traders' valuation of the stock may be revised. An increase (decrease) in the stock price may cause the traders to revise their valuation upward, and may result in the cancellation of their seemingly stale limit buy (sell) orders. The greater the magnitude of the change, the higher will be the probability of cancellation. The *bidsz* and *ofrsz* are the bid and offer size (the depth at the best quotes in thousands of shares) respectively. For the case of a limit buy order submitted at the bid, a larger bid size after the order is submitted will not affect the probability of its execution. However, if the limit buy order

is submitted behind the best bid, a larger bid size will lengthen the time needed for an execution, and also reduce the probability of its execution. Hence, a larger bid size after the submission of a limit buy order should increase the probability of its cancellation. A larger offer size, on the other hand, will encourage the use of market sell orders instead of limit sell orders. The increase of the use of market sell orders will in turn increase the probability of execution of limit buy orders, and hence reduce the probability of their cancellation. Therefore, the sign of the coefficient for *bidsz* and *ofrsz* should be positive and negative for limit buy orders respectively. The reverse should be true for limit sell orders.

Table IX shows the results for the probit analysis. Panel A and B show the results for limit buy and sell orders respectively. The direction of change of the bid and offer prices does have an impact on the decision of traders to cancel their limit orders. A positive change in the bid increases the probability of cancellation of limit buy orders, and a negative change in the offer increases the probability of cancellation for limit sell orders. Contrary to expectation, larger order sizes do not increase the probability of cancellation of limit orders but instead decreases probability significantly. A possible explanation for this may be that traders realize that larger orders are exposed to larger risk, and hence are more cautious in pricing the orders, resulting in more execution than cancellation. An increase in the spread, after order submission, does increase the probability of cancellation for both the limit buy and sell orders. However, the positive relationship is not significant for the 300- and 600-second time windows. Other than for the 10- and 30-second time windows in Panel A, and for the 10-second window in Panel B, limit orders

are significantly more likely to be cancelled as they are submitted later in the day. However, orders that are submitted during the last half hour of trading are less likely to be cancelled. The coefficients of *atime* are negative and significant for all the time windows. An increase in the trading price of the stock increases the probability of cancellation for limit buy orders for the 120-, 300-, and 600-second time windows. However, only the coefficient in the 300-second window is significant. As for limit sell orders, though the coefficients for *cngpx* have the right sign for the shorter four time windows, only the one for the 120-second window is significant. Larger bid (offer) sizes do not seem to encourage cancellation of limit buy (sell) orders. In fact, cancellation is less likely especially for the shorter time windows. Nevertheless, larger offer (bid) sizes do reduce the probability of cancellation for limit buy (sell) orders, probably because they attract more market sell (buy) orders over limit sell orders.

VI. Conclusion

This paper focuses on the large number of cancellation orders present in the NYSE today. Evidence in the sample shows that cancellations are used by limit order traders as part of their trading strategies. Cancellations as a result of the three trading strategies examined in this paper, namely the order splitting strategy, the undercutting strategy, and the limit order replacement strategy, constitute up to 85% of the total number of cancellations in the sample. I also show that most traders exit the market, that is they stop pursuing their interest in a particular stock, at least for that particular day, upon canceling their orders for that stock. But if they choose to resubmit their orders, they are more likely to resubmit

them as limit orders with more aggressive limit prices than as market orders. The probit analysis shows the variables that may influence a trader's decision to leave his orders alone or to cancel it. The probability of cancellation of a limit order is positively related to the time of the day and the change in the spread. Orders that are submitted during the last half hour of trading have lower probability of cancellation. Limit buy (sell) orders are more likely to be cancelled when the bid (offer) and the trading price increase (decrease).

The results of this paper have at least two significant implications on the current literature, which may justify future research. First, results of studies examining the performance of limit orders vs market orders, which assume that failed limit orders are immediately resubmitted as market orders, may be biased. Most traders either do nothing after they cancel their orders or resubmit them as limit orders. Future performance analysis of limit orders vs market orders will have to take into account such a finding. Second, the market has witness an increase in the proportion of limit orders relative to market orders over the last decade. However, this paper shows that more than 40% of all limit orders are cancelled, and more than 95% of the cancellations occur within 10 minutes of submission. Under such findings, the increase in liquidity provided by the increase in the proportion of limit orders is questionable. However, a related and more important question will be the impact of such a trend of cancellation on the price discovery process.

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Footnotes:

[1] Most of the limit orders are day orders. Both the day orders and good-to-cancel orders are included in the sample. As a robustness check, all the analysis in this paper was repeated after removing all the good-to-cancel orders. The results are almost identical.

[2] Even by lifting any time constraint between cancellation and resubmission (i.e. extending the time to replace to the end of the trading day), only up to 44.08% of all cancellations are replaced by either market orders or more aggressively priced limit orders.

Table I: Distribution of Orders Submitted

This table shows the distribution of the submission of the different types of orders, including cancellation orders. The different types of orders are market buy (MB), market sell (MS), limit buy (LB), limit sell (LS), and cancellation (CXL). Column 1 gives the number of each type of orders submitted, while column 2 shows the number as a percentage of the total number of orders submitted.

Total Number of Orders= 744,969

Type of Orders	Number of Orders	% of Total Orders
MB	90,413	12.14
MS	106,275	14.27
LB	202,397	27.17
LS	180,164	24.18
CXL	165,720	22.25

Table II: Descriptive Statistics of the Cancellations

This table shows the break down of the cancellation orders. Panel A breaks down the cancellations into market orders (MO) or limit orders (LO). Panel B gives the number and percentage of the cancellation of limit orders that are matched with the originating orders. Panel C shows the number and percentage of the limit buy and sell orders, among the originating orders of the cancellations. Panel D shows the break down of the matched cancellations of the limit buy and limit sell orders into whether the cancellations occur when there is a negative, no, or positive change in the bid and offer price respectively.

Total Number of Cancellations= 165,720

Panel A:

Number of cancellations of MO	2,866	1.73%
Number of cancellations of LO	162,854	98.27%

Panel B:

Number of cancellations of LO matched	145,572	89.39%
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Panel C:

Number of Matched LB	75,710	52.01%
Number of Matched LS	69,862	47.99%

Panel D:

LB:

Negative change in bid	11,854	15.66%
No change in bid	26,580	35.11%
Positive change in bid	37,276	49.24%

LS:

Negative change in Offer	38,407	54.98%
No change in Offer	22,971	32.88%
Positive change in Offer	8,484	12.14%

Table III: Distribution of the Time Interval between Limit Order Submission and Cancellation

This table shows the distribution of the time interval between the submission of the limit orders and their cancellations. Column 1 shows the number of cancellations that falls into the respective time bins. Column 2 shows the figures in Column 1 as a percentage of the total number of cancellations, while Column 3 shows the cumulative percentage.

Total no. of cancellations= **145,572**

Time (sec)	No. of Obs	% of Obs	Cum. % of Obs
0	1,574	1.08	1.08
1 - 10	29,719	20.42	21.50
11 - 30	32,906	22.60	44.10
31 - 60	25,941	17.82	61.92
61 - 300	42,247	29.02	90.94
300 - 600	6,614	4.54	95.49
> 600	6,571	4.51	100.00

Table IV: Order Splitting Strategy and Cancellations

This table shows the percentage of cancellations caused by the order splitting strategy. Cancellations, with originating orders of the same type (market buy, market sell, limit buy, and limit sell), for the same stock, submitted by the same branch of the same firm, within a specific time interval, are considered to be caused by the order splitting strategy. The time windows considered are 10, 30, 60, 120, 300 seconds. Column 1 shows the cancellation of split orders as a percentage of the total number of cancellations in the full sample. The full sample is broken down into quartiles according to trading activity: Quartile 1 contains the most active stocks while Quartile 4 contains the least active ones. Columns 2 through 5 show the percentage of the cancellation of order splits for Quartile 1 through 4 respectively.

Total no. of cancellations= 145,572

Time Window (secs)	Full	Q1	Q2	Q3	Q4
10	12.67	13.94	9.72	4.04	0.84
30	29.19	31.72	23.78	6.56	1.68
60	43.36	46.49	37.11	10.35	5.04
120	56.96	60.16	50.99	19.01	7.56
300	71.49	74.32	66.78	31.65	12.61

Table V: Undercutting Strategy and Cancellations

This table shows the percentage of cancellations caused by the undercutting strategy. The cancellation of limit orders that improve on the best quotes, within a specific time interval, after they have just been improved (by other limit orders), are considered to be cancellations caused by the undercutting strategy. The time windows considered are 10, 30, 60, 120, 300 seconds. Column 1 shows the cancellation of undercutting limit orders as a percentage of the total number of cancellations in the full sample. The full sample is broken down into quartiles according to trading activity: Quartile 1 contains the most active stocks while Quartile 4 contains the least active ones. Columns 2 through 5 show the percentage of the cancellation of undercutting limit orders for Quartile 1 through 4 respectively.

Total no. of cancellations= 145,572

Time Window (secs)	Full	Q1	Q2	Q3	Q4
-					
10	19.21	20.36	16.78	8.79	3.36
30	22.76	23.03	22.47	17.43	6.72
60	23.78	23.48	24.54	25.51	15.97
120	24.29	23.60	25.63	32.82	19.33
300	24.59	23.61	26.12	40.41	31.09

Table VI: Replacement of Cancellations by Market Orders

This table shows the percentage of cancelled limit orders that are replaced by market orders. Market orders of the same direction (buy or sell), submitted by the same branch of the same firm, within a specific time interval from the time of cancellation of the limit orders, are considered to be replacement market orders. The time windows considered are 10, 30, 60, 120, 300 seconds. Panel A shows the results with the added condition that the order size of the cancelled limit orders and their replacement market orders are the same, while Panel B relaxes this constraint. Column 1 shows the results for the full sample. The full sample is broken down into quartiles according to trading activity: Quartile 1 contains the most active stocks while Quartile 4 contains the least active ones. Columns 2 through 5 show the results for Quartile 1 through 4 respectively.

Total no. of cancellations= 145,572

Time Window (secs)	Full	Q1	Q2	Q3	Q4
<u>Panel A:</u>					
10	0.11	0.13	0.04	0.06	0.00
30	0.21	0.26	0.09	0.06	0.00
60	0.31	0.37	0.15	0.10	0.00
120	0.39	0.46	0.19	0.13	0.00
300	0.49	0.59	0.25	0.19	0.00
<u>Panel B:</u>					
10	0.24	0.30	0.07	0.16	0.00
30	0.49	0.61	0.19	0.16	0.00
60	0.71	0.86	0.30	0.23	0.00
120	0.93	1.13	0.40	0.32	0.00
300	1.26	1.53	0.54	0.48	0.00

Table VII: Replacement of Cancellations by Limit Orders

This table shows the percentage of cancelled limit orders that are replaced by more aggressively priced limit orders. Limit orders of the same direction (buy or sell), submitted by the same branch of the same firm, with more aggressive prices, within a specific time interval from the time of cancellation of the limit orders, are considered to be replacement limit orders. The time windows considered are 10, 30, 60, 120, 300 seconds. Panel A shows the results with the added condition that the order size of the cancelled limit orders and their replacement limit orders are the same, while Panel B relaxes this constraint. Column 1 shows the results for the full sample. The full sample is broken down into quartiles according to trading activity: Quartile 1 contains the most active stocks while Quartile 4 contains the least active ones. Columns 2 through 5 show the results for Quartile 1 through 4 respectively.

Total no. of cancellations= 145,572

Time Window (secs)	Full	Q1	Q2	Q3	Q4
<u>Panel A:</u>					
10	7.22	7.18	7.65	3.94	3.36
30	10.94	11.08	10.90	6.72	6.72
60	12.70	12.82	12.61	9.54	7.56
120	14.10	14.23	13.98	11.28	11.76
300	15.88	16.00	15.66	14.35	12.61
<u>Panel B:</u>					
10	13.85	13.82	14.69	5.11	5.88
30	21.94	22.38	21.73	9.76	9.24
60	25.87	26.41	25.38	13.71	10.08
120	29.51	30.12	28.87	16.62	14.29
300	34.19	34.81	33.54	21.34	15.13

Table VIII: Order Splitting Strategy, Undercutting Strategy, Limit Order Replacement Strategy and Cancellations

This table shows the percentage of cancellations caused by the order splitting strategy, undercutting strategy, and limit order replacement strategy. Cancellations, with originating orders of the same type (market buy, market sell, limit buy, and limit sell), for the same stock, submitted by the same branch of the same firm, within a specific time interval, are considered to be caused by the order splitting strategy. The cancellation of limit orders that improve on the best quotes, within a specific time interval, after they have just been improved (by other limit orders), are considered to be cancellations caused by the undercutting strategy. Cancellations, which are followed by the submission of limit orders of the same direction (buy or sell), submitted by the same branch of the same firm, with more aggressive prices, within a specific time interval from the time of cancellation of the limit orders, are considered to be caused by the limit order replacement strategy. Panel A shows the results with the added condition that the order size of the cancelled limit orders and their replacement limit orders are the same, while Panel B relaxes this constraint. Column 1 shows the results for the full sample. The full sample is broken down into quartiles according to trading activity: Quartile 1 contains the most active stocks while Quartile 4 contains the least active ones. Columns 2 through 5 show the results for Quartile 1 through 4 respectively.

Total no. of cancellations= 145,572

Time Window (secs)	Full	Q1	Q2	Q3	Q4
<u>Panel A:</u>					
10	33.78	35.84	29.41	15.45	6.72
30	50.35	52.62	45.78	27.09	12.61
60	61.56	63.84	57.06	37.37	23.53
120	71.50	73.64	67.30	48.59	30.25
300	81.96	83.79	78.43	62.11	41.18
<u>Panel B:</u>					
10	38.18	40.19	34.21	16.42	9.24
30	55.64	57.90	51.40	29.13	15.13
60	66.12	68.33	62.01	39.86	26.05
120	75.38	77.41	71.62	51.24	32.77
300	85.03	86.68	82.06	64.99	42.86

Table IX: Probit Analysis of the Probability of Cancellation of Limit Orders

This table presents the results of the probit analysis for the probability of cancellation of the limit orders. Panels A and B show the results for the limit buy and limit sell orders respectively. The dependent variable is the dummy variable that equals 1, if the limit order is cancelled within a specific time interval after it is submitted, and 0 otherwise. Six different time windows are tested: 10, 30, 60, 120, 300, and 600 seconds. The independent variables are *cngbid* (dummy variable that equals 1, if there is a positive change in the bid within the time window, and 0 otherwise), *cngofr* (dummy variable that equals 1, if there is a negative change in the offer within the time window, and 0 otherwise), *ordsz* (order size), *spread* (percentage change in the spread within the time window), *otime* (order time, in number of hours after the market opens), *atime* (dummy variable that equals 1, if the order is submitted half an hour before market closes, and 0 otherwise), *cngpx* (percentage change in the price of the stock within the time window), *bidsz* (bid size), and *ofrsz* (offer size). Columns 1 through 6 show the slope of the coefficients for the probit regression for each of the time windows. Entries marked with “*”, “**”, and “***” represent coefficients that are significant at the 10%, 5% and 1% level.

	10 sec		30 sec		60 sec		120 sec		300 sec		600 sec	
<u>Panel A: Limit Buy Orders</u>												
cngbid	0.0131	***	0.0149	***	0.0119	***	0.0142	***	0.0060	***	0.0097	***
ordsz	-0.0015	***	-0.0027	***	-0.0043	***	-0.0035	***	-0.0034	***	-0.0008	***
spread	0.0030	***	0.0025	***	0.0015	***	0.0025	***	0.0006		0.0002	
otime	0.0005		0.0004		0.0014	***	0.0015	***	0.0020	***	0.0042	***
atime	-0.0149	***	-0.0094	***	-0.0178	***	-0.0204	***	-0.0291	***	-0.0409	***
cngpx	-0.5751		-0.1235		-0.0127		0.1239		0.2767	***	0.1881	
bidsz	-0.0425	***	-0.0645	***	-0.0382	**	-0.0236		-0.0117		-0.0247	
ofrsz	-0.0535	***	-0.0432	***	-0.0493	***	-0.0424	***	-0.0410	**	0.0217	
<u>Panel B: Limit Sell Orders</u>												
cngofr	0.0136	***	0.0214	***	0.0163	***	0.0141	***	0.0069	***	0.0150	***
ordsz	-0.0009	***	-0.0040	***	-0.0040	***	-0.0046	***	-0.0025	***	-0.0015	***
spread	0.0041	***	0.0034	***	0.0019	***	0.0014	***	0.0007		0.0003	
otime	0.0001		0.0014	***	0.0015	***	0.0033	***	0.0048	***	0.0065	***
atime	-0.0091	***	-0.0153	***	-0.0133	***	-0.0214	***	-0.0020		-0.0046	
cngpx	-0.1657		-0.0202		-0.1552		-0.1978	*	0.0310		0.0107	
bidsz	-0.0598	***	-0.0479	***	-0.0849	***	-0.0436	***	-0.0306	*	-0.0498	**
ofrsz	-0.0490	***	-0.0466	***	-0.0348	**	-0.0505	***	-0.0159		-0.0093	